

확률론 (MAT4033)

Lecture notes for Probability Theory (2026 Fall)

김민현 교수님

작성 : 공과대학 산업공학과 박성욱

Last updated: September 29, 2026

Contents

1. Review of Probability	1
1.1 Events and Probability	1
1.2 Random Variables	4
1.3 Conditional Probability and Independence	7
2. Conditional Expectation	9
2.1 Conditioning on a Event	9
2.2 Conditioning on a discrete random variable	9
2.3 Conditioning on an arbitrary random variable	11
2.4 Conditioning on a σ -field	13
2.5 General Properties	14
3. Martingales in Discrete Time	17
3.1 Sequences of Random Variables	17
3.2 Filtrations	18
3.3 Martingales	19
3.4 Games of Chance	21
Appendix	22
A. Measure Theory	22
A.1 Measures and Measurable Functions	22

Chapter 1. Review of Probability

1.1 Events and Probability

In this course, we study the probability in the sense of real analysis, which relies on the measure theory. Hence, we will translate the notions of real analysis to the language of probability theory. Now we start with some definitions.

Definition 1.1. Let $\Omega \neq \emptyset$. A family of

$$\mathcal{F} \subset 2^\Omega = \mathcal{P}(\Omega)$$

is a σ -field if it satisfies the following properties:

1. $\emptyset \in \mathcal{F}$.
2. If $A \in \mathcal{F}$, then $\Omega \setminus A \in \mathcal{F}$.

3. If $A_1, A_2, A_3, \dots \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$.

Note that σ -field is nothing but a collection of subsets¹ that is closed under the complement and countable union, which contains the empty set. From now on, we will denote $\mathcal{F}$ as a σ -field on Ω . Based on the above definition, we can easily observe the following remarks.

Remark. Ω and $\mathcal{F}$ are defined as above.

1. $\Omega \in \mathcal{F}$. Use the first two properties to show this.
2. $\bigcap_{n=1}^{\infty} A_n \in \mathcal{F}$. Use the second and third properties to show this.
3. If $A, B \in \mathcal{F}$, then $A \setminus B \in \mathcal{F}$.

Example. The example of σ -fields are as follows.

1. $2^{\mathbb{R}}$ is a σ -field.
2. (Borel σ -field) Recall that we have defined the Borel σ -field $\mathcal{B}(\mathbb{R})$ in the real analysis class. We say that $\mathcal{B}(\mathbb{R})$ is **Borel σ -field** which is the smallest σ -field containing all open sets² in $\mathbb{R}$. Its elements are called **Borel sets**.
3. The family of (Lebesgue) measurable sets $\mathcal{M}(\mathbb{R})$ is a σ -field.
4. (Coin Tossing) To see more practical examples, consider the following example. For two coin tosses, take

$$\Omega = \{HH, HT, TH, TT\}.$$

If only the first coin toss is observed, let

$$A = \{HH, HT\}, \quad B = \{HH, TH\}.$$

Then the available information is described by the σ -field,

$$\mathcal{F} = \{\emptyset, A, \Omega \setminus A, \Omega\}.$$

Definition 1.2. A **probability measure** is a map

$$P : \mathcal{F} \rightarrow [0, 1]$$

such that

1. $P(\Omega) = 1$.
2. (Countable Additivity) For pairwise disjoint sets $A_n \in \mathcal{F}$,

$$P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n).$$

We call Ω the **sample space**, $\mathcal{F}$ the **event space**, and $A \in \mathcal{F}$ the **event**. The triple $(\Omega, \mathcal{F}, P)$ is called a **probability space**. An event A is said to be hold **almost surely (a.s.)** if $P(A) = 1$.

Note that $P(A) = 1$ does not imply that $A = \Omega$.

¹That is, the power set of underlying set Ω .

²For the real line $\mathbb{R}$, all open sets mean all open intervals of $\mathbb{R}$.

Example. Take $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$ and $P = \text{Leb}([0, 1])^3$. Indeed,

$$P(\Omega) = P([0, 1]) = \text{Leb}([0, 1]) = 1$$

and more generally, for $[a, b] \subset [0, 1]$,

$$P([a, b]) = \text{Leb}([a, b]) = b - a.$$

Therefore we can observe that such $(\Omega, \mathcal{F}, P)$ is a probability space.

Remark. Let $(\Omega, \mathcal{F}, P)$ be a probability space.

1. $P(\Omega) = 1$.
2. $P(\Omega \setminus A) = 1 - P(A)$ for any $A \in \mathcal{F}$.
3. $A \subset B \implies P(A) \leq P(B)$.
4. (Countable subadditivity) $P(\bigcup_{n=1}^{\infty} A_n) \leq \sum_{n=1}^{\infty} P(A_n)$.

The countable *subadditivity* shows that even though we do not have *pairwise* disjoint sets, we still get *inequality*.

Let us define some notations about limit of sets.

If $E_j \subset E_{j+1}$ and $\bigcup_{j=1}^{\infty} E_j = E$, we write $E_j \nearrow E$.

If $E_j \supset E_{j+1}$ and $\bigcap_{j=1}^{\infty} E_j = E$, we write $E_j \searrow E$.

For the measurable sets, we studied that the following corollary holds. Note that $m(\cdot)$ denotes the Lebesgue measure on $\mathbb{R}$ for this moment.

Corollary 3.3 (From Stein's book). Let $E_j \in \mathcal{M}$ for all $j \in \mathbb{N}$.

1. If $E_j \nearrow E$, then $\lim_{j \rightarrow \infty} m(E_j) = m(E)$.
2. If $E_j \searrow E$ and $m(E_k) < \infty$ for some k , then $\lim_{j \rightarrow \infty} m(E_j) = m(E)$.

Similarly, we can obtain the followings for the probability measure P but see the difference with the Lebesgue measure.

Exercise 1.1 (Continuity of Probability). Let $A_i \in \mathcal{F}$.

1. If $A_n \nearrow A$, then $\lim_{n \rightarrow \infty} P(A_n) = P(A)$.
2. If $A_n \searrow A$, then $\lim_{n \rightarrow \infty} P(A_n) = P(A)$.

Remarkably, we can drop the finite condition in the second part, which is different from the Lebesgue measure since $P(\Omega) = 1 < \infty$.

Again, let us compare the Borel-Cantelli lemma for the Lebesgue measure and the probability measure. In the case of the Lebesgue measure, this lemma implies that if we sum up countably many sets, and if its sum is finite, then the *tail* of the sets have measure zero. Similar happens for the probability measure, which is stated as follows.

Exercise 1.6.16 (The Borel-Cantelli Lemma From Stein's book). Suppose $\{E_k\}_{k=1}^{\infty}$ is a countable family of measurable subsets of $\mathbb{R}^d$ and that

$$\sum_{k=1}^{\infty} m(E_k) < \infty.$$

Let

$$E = \{x \in \mathbb{R}^d : x \in E_k \text{ for infinitely many } k\} = \limsup_{k \rightarrow \infty} E_k.$$

Then, $E \in \mathcal{M}$ and $m(E) = 0$.

³Here, $\text{Leb}([0, 1])$ denotes the Lebesgue measure on $[0, 1]$.

Lemma 1.1 (Borel-Cantelli). If $\sum_{n=1}^{\infty} P(A_n) < \infty$ and $B_n = \bigcup_{k \geq n} A_k$, then

$$P\left(\bigcap_{n \geq 1} B_n\right) = 0.$$

Proof. Note that

$$P(B_n) \leq \sum_{k \leq n} P(A_k) \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Since $B_n \searrow \bigcap_{n \geq 1} B_n$, we have

$$P\left(\bigcap_{n \geq 1} B_n\right) = \lim_{n \rightarrow \infty} P(B_n) = 0.$$

□

1.2 Random Variables

Start with the definition of random variables.

Definition 1.3. A function $\xi : \Omega \rightarrow \mathbb{R}$ is **$\mathcal{F}$ -measurable**⁴ if for all Borel sets $B \in \mathcal{B}(\mathbb{R})$,

$$\{\xi \in B\} := \{\omega \in \Omega : \xi(\omega) \in B\} = \xi^{-1}(B) \in \mathcal{F}.$$

If $(\Omega, \mathcal{F}, P)$ is a probability space, then such a map ξ is a **random variable**.

Definition 1.4. The σ -field generated by random variable ξ is defined by

$$\sigma(\xi) := \{\{\xi \in B\} : B \in \mathcal{B}(\mathbb{R})\}.$$

Note that $\sigma(\xi)$ contains all the information from observing ξ .

Definition 1.5. For a family $\{\xi_i : i \in \mathcal{I}\}$,

$$\sigma(\xi_i : i \in \mathcal{I})$$

is the smallest σ -field containing every event $\{\xi_i \in B\}$.

Exercise 1.3. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is Borel (measurable)⁵, then $f(\xi)$ is $\sigma(\xi)$ -measurable.

Proof. Trivial. □

Surprisingly, the converse of the Exercise 1.3. is also true. This work is done by Doob and Dynkin, the great mathematicians who contributed to the development of probability theory.

Lemma 1.2 (Doob-Dynkin). If η is $\sigma(\xi)$ -measurable, then

$$\eta = f(\xi) \text{ for some Borel function } f : \mathbb{R} \rightarrow \mathbb{R}.$$

To prove this lemma, we have to construct a such Borel function f . The construction of such f is highly non-trivial, thereby we omit the proof of this lemma. It is enough to see the statement for this time.

⁴Note : Let $E \subset \mathbb{R}^d$. A function $f : E \rightarrow \mathbb{R}^d$ is measurable if for all $a \in \mathbb{R}$, the set

$$\{f < a\} = f^{-1}(\{x \in \mathbb{R} : x < a\}) = \{x \in E : f(x) < a\} \in \mathcal{M}.$$

Thus the measurability is closely related to the inverse image of the function.

⁵Our textbook defines **Borel function** as follows. A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is Borel if the inverse image $f^{-1}(B)$ of any Borel set B is a Borel set. In other words, f is Borel if it is measurable with respect to the Borel σ -field.

Definition 1.6. Every random variable ξ gives rise to a probability measure

$$P_\xi(B) = P(\{\xi \in B\}) \quad \text{for } B \in \mathcal{B}(\mathbb{R}).$$

P_ξ is called the **distribution** of ξ . The function $F_\xi(x) : \mathbb{R} \rightarrow [0, 1]$ defined by

$$F_\xi(x) = P_\xi((-\infty, x]) = P(\{\xi \leq x\})$$

is called the **distribution function** of ξ .

Exercise 1.4. F_ξ is non-decreasing, right-continuous, and satisfies

$$\lim_{x \rightarrow -\infty} F_\xi(x) = 0, \quad \text{and} \quad \lim_{x \rightarrow \infty} F_\xi(x) = 1.$$

Proof.

1. Monotonicity is immediate.

2. If $x_n \searrow x$, then $\{\xi \leq x_n\} \searrow \{\xi \leq x\}$ hence $F_{x_n} \rightarrow F_\xi(x)$.

3. For the limits, use

$$\{\xi \leq -n\} \searrow \emptyset \quad \text{and} \quad \{\xi \leq n\} \nearrow \Omega.$$

□

Definition 1.7. The law of ξ is **absolutely continuous**⁶ if there is a Borel function $f_\xi : \mathbb{R} \rightarrow \mathbb{R}$ such that for any Borel set $B \in \mathcal{B}(\mathbb{R})$,

$$P_\xi(B) = P(\xi \in B) = \int_B f_\xi(x) dx$$

and f_ξ is called the **density** of ξ . The law is **discrete** if there exists at most countable pairwise distinct values $\{x_i\}_{i \in \mathcal{I}}$ such that

$$P(\xi \in B) = \sum_{x_i \in B} P(\xi = x_i) \quad \text{for all } B \in \mathcal{B}(\mathbb{R}).$$

Exercise 1.5 and 1.6. For a continuous density f_ξ , we have

$$F'_\xi(x) = f_\xi(x).$$

For a discrete law,

F_ξ jumps by $P(\xi = x_i)$ at each x_i

Proof. The disjoint atoms in $(s, t]$ give

$$F_\xi(t) - F_\xi(s) = P\{s < \xi \leq t\} = \sum_{s < x \leq t} P\{\xi = x\}.$$

Thus, the increment is zero if there is no atom in $(s, t]$, and

$$F_\xi(x_i) - F_\xi(x_i-) = P\{\xi = x_i\}.$$

□

⁶That is, the distribution of ξ is absolutely continuous.

Definition. The joint distribution of

$$\xi_1, \xi_2, \dots, \xi_n$$

is a probability measure defined by

$$P_{\xi_1, \xi_2, \dots, \xi_n}(B) = P\{(\xi_1, \xi_2, \dots, \xi_n) \in B\}, \quad B \in \mathcal{B}(\mathbb{R}^n).$$

Definition. $\xi \in L^1(\Omega, \mathcal{F}, P)$ if

$$\int_{\Omega} |\xi| dP < \infty.$$

Then

$$E(\xi) = \int_{\Omega} \xi dP$$

exists and is called the expectation of ξ .

Example of computing expectation.

1. For $A \in \mathcal{F}$, $E(\mathbb{1}_A) = P(A)$.
2. If η is a step function, that is,

$$\eta = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$$

with pairwise disjoint $A_i \in \mathcal{F}$, then

$$E(\eta) = \sum_{i=1}^n a_i P(A_i).$$

Exercise 1.7 (Law of the Unconscious Statistician(LOTUS)). If $h : \mathbb{R} \rightarrow \mathbb{R}$ is a Borel measurable function and $h(\xi)$ is integrable, then

$$E(h(\xi)) = \int_{\mathbb{R}} h(x) dP_{\xi}(x).$$

Hence,

1. $E(h(\xi)) = \int_{\mathbb{R}} h(x) dP_{\xi}(x)$ for a continuous law
2. $E(h(\xi)) = \sum_{i=1}^{\infty} h(x_i) P\{\xi = x_i\}$ for a discrete law.

Definition (Variance). $\xi \in L^2(\Omega, \mathcal{F}, P) \subset L^1$ if

$$E(|\xi|^2) = \int_{\Omega} |\xi|^2 dP < \infty.$$

Its variance is

$$\text{var}(\xi) = E((\xi - E(\xi))^2) = E(\xi^2) - (E(\xi))^2.$$

Exercise 1.8 (Cauchy-Schwarz inequality).

$$|E(\xi\eta)|^2 \leq E(\xi^2)E(\eta^2).$$

Consequently, $L^2 \subset L^1$, and

$$E(|\xi|) \leq \sqrt{E(\xi^2)}.$$

Proof. Let $\xi, \eta \in L^2$. Then

$$\xi\eta \in L^1 \quad \text{since} \quad 2|\xi\eta| \leq \xi^2 + \eta^2.$$

The quadratic function

$$t \mapsto E((\xi - t\eta)^2) = E(\xi^2) - 2tE(\xi\eta) + t^2E(\eta^2) \quad \text{is non negative.}$$

So its discriminant is non positive:

$$E(\xi\eta)^2 - E(\xi^2)E(\eta^2) \leq 0 \implies E(\xi\eta)^2 \leq E(\xi^2)E(\eta^2).$$

Apply this to ξ and $\mathbb{1}$ to

$$E(|\xi|) \leq \sqrt{E(\xi^2)}.$$

□

Excercise 1.9 (Coarea formula). If $\eta \geq 0$, then

$$E(\eta^2) = 2 \int_0^\infty tP\{\eta > t\}dt.$$

Proof. We have⁷

$$\eta^2 = \int_0^\infty 2t\mathbb{1}_{\{\eta > t\}}dt.$$

Tonelli's theorem gives

$$\begin{aligned} E(\eta^2) &= \int_\Omega \int_0^\infty 2t\mathbb{1}_{\{\eta > t\}}dt dP \\ &= \int_0^\infty 2tE(\mathbb{1}_{\{\eta > t\}})dt \\ &= \int_0^\infty 2tP\{\eta > t\}dt. \end{aligned}$$

□

1.3 Conditional Probability and Independence

Definition. Let $A, B \in \mathcal{F}$ be such that $P(B) > 0$. The conditional probability of A given B is defined by

$$P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

Excercise 1.10. If $B_1, \dots$ are pairwise disjoint, $\bigcup_{n \geq 1} B_n = \Omega$, and $P(B_n) > 0$, then

$$P(A) = \sum_{n \geq 1} P(A|B_n)P(B_n).$$

Proof. Since $A = \bigcup_{n \geq 1} (A \cap B_n)$, we have

$$P(A) = \sum_{n \geq 1} P(A \cap B_n) = \sum_{n \geq 1} P(A|B_n)P(B_n).$$

□

⁷Note that $\int_0^\eta 2t\mathbb{1}_{\{\eta > t\}}dt \rightarrow \eta^2$ and $\int_\eta^\infty 2t\mathbb{1}_{\{\eta > t\}}dt \rightarrow 0$ since $\mathbb{1}_{\{\eta > t\}}$ is 1 when $\eta > t$ and 0 otherwise.

Remark (Bayes' rule). If $P(A) > 0$, then

$$P(B_j|A) = \frac{P(A|B_j)P(B_j)}{\sum_{n \geq 1} P(A|B_n)P(B_n)}.$$

Definition. Events A and B are independent if

$$P(A \cap B) = P(A)P(B).$$

Events $A_1, \dots, A_n$ are independent if for every subfamily,

$$P(A_{i_1} \cap \dots \cap A_{i_k}) = P(A_{i_1}) \dots P(A_{i_k}).$$

Exercise 1.11. If $P(B) > 0$, then

$$A \text{ and } B \text{ are independent} \iff P(A|B) = P(A).$$

Definition. Random variables η, ξ are independent if

$$P(\eta \in A, \xi \in B) = P(\eta \in A)P(\xi \in B)$$

for all $A, B \in \mathcal{B}(\mathbb{R})$. An arbitrary family is independent if every finite subfamily is independent.

Proposition 1.1. If $\xi_1, \dots, \xi_n$ are independent, then

$$E(\xi_1 \dots \xi_n) = \prod_{i=1}^n E(\xi_i).$$

In particular, independent $\xi, \eta \in L^2$ are uncorrelated:

$$E(\eta\xi) = E(\eta)E(\xi)$$

Definition. Two σ -fields $\mathcal{G}, \mathcal{H} \in \mathcal{F}$ are independent if for every $A \in \mathcal{G}, B \in \mathcal{H}$,

$$P(A \cap B) = P(A)P(B).$$

Similarly, $\mathcal{G}_1, \dots, \mathcal{G}_n$ are independent if $A_1, \dots, A_n$ are independent whenever $A_i \in \mathcal{G}_i$.

Exercise 1.12. Random variables η, ξ are independent if and only if the σ -fields $\sigma(\eta), \sigma(\xi)$ are independent.

Proof. Every event $A \in \sigma(\xi)$ has the form

$$A = \{\xi \in C\} \quad \text{for some } C \in \mathcal{B}(\mathbb{R}).$$

and $B \in \sigma(\eta)$ has the form

$$B = \{\eta \in D\} \quad \text{for some } D \in \mathcal{B}(\mathbb{R}).$$

Hence independence of two σ -fields means

$$P(\xi \in C, \eta \in D) = P(\xi \in C)P(\eta \in D)$$

□

Definition. A random variable ξ is independent of a σ -field $\mathcal{G}$ if

$$\sigma(\xi) \text{ and } \mathcal{G} \text{ are independent.}$$

A mixed family of random variables and σ -fields are independent if replacing each random variable by its generated σ -field produces an independent family of σ -fields.

2. Conditional Expectation

2.1 Conditioning on a Event

Definition. Let $\xi \in L^1$ and let $B \in \mathcal{F}$ with $P(B) > 0$. Then the conditional expectation of ξ given B is defined by

$$E(\xi|B) = \frac{1}{P(B)} \int_B \xi dP.$$

Example 2.1. Fair 10p, 20p, and 50p coins are tossed. Let ξ be the value of the coins showing heads, and let B be the event that exactly two coins show heads. Then

$$B = \{HHT, HTH, THH\}, \quad \text{and} \quad E(\xi|B) = \frac{1}{\frac{3}{8}} \left(\frac{30}{8} + \frac{60}{8} + \frac{70}{8} \right) = \frac{160}{3}.$$

Exercise 2.1.

$$E(\xi|\Omega) = E(\xi)$$

Proof.

$$E(\xi|\Omega) = \frac{1}{P(\Omega)} \int_{\Omega} \xi dP = E(\xi)$$

□

Exercise 2.2. Show that if

$$\mathbb{1}_A(\omega) = \begin{cases} 1, & \omega \in A \\ 0, & \omega \notin A \end{cases}$$

then

$$E(\mathbb{1}_A|B) = P(A|B) \quad \text{where} \quad P(A|B) = \frac{P(A \cap B)}{P(B)}.$$

2.2 Conditioning on a discrete random variable

Definition. Let $\xi \in L^1$, and let η be a discrete random variable, that is η takes distinct values

$$y_1, \dots, y_n \quad \text{with} \quad P(\eta = y_i) > 0 \quad \text{for } i = 1, \dots, n.$$

Define

$$E(\xi|\eta)(\omega) := E(\xi|\{\eta = y_n\}) \quad \text{if} \quad \eta(\omega) = y_n.$$

Equivalently,

$$E(\xi|\eta)(\omega) = \sum_{n \geq 1} E(\xi|\{\eta = y_n\}) \mathbb{1}_{\{\eta = y_n\}}(\omega).$$

Example 2.2. Consider the coin tossing example as in Example 2.1. Let ξ be the total value of the coins showing heads and let η be the contribution from 10p and 20p coins.

If $\eta = 0$, only the 50p coin remains unknown, so

$$E(\xi|\{\eta = 0\}) = \frac{1}{\frac{1}{4}} \left(\frac{50}{8} + \frac{0}{8} \right) = 25.$$

Keep calculating for $\eta = 10, 20, 30$, we have

η	0	10	20	30
$E(\xi \eta)$	25	35	45	55

which implies that $E(\xi|\eta) = \eta + 25$.

Example 2.3. Take $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$, and P be the Lebesgue measure $m(\cdot)$. Let $\xi(x) = 2x^2$ and

$$\eta(x) = \begin{cases} 0, & 0 \leq x \leq \frac{1}{3} \\ 1, & \frac{1}{3} < x \leq \frac{2}{3} \\ 2, & \frac{2}{3} < x \leq 1 \end{cases}$$

For $x \in [0, \frac{1}{3}]$,

$$E(\xi|\eta)(x) = \frac{1}{\frac{1}{3}} \int_0^{\frac{1}{3}} 2t^2 dt = \frac{2}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} = \frac{2}{27}.$$

The result is,

$$E(\xi|\eta)(x) = \begin{cases} \frac{2}{27}, & 0 \leq x \leq \frac{1}{3} \\ \frac{14}{27}, & \frac{1}{3} < x \leq \frac{2}{3} \\ \frac{38}{27}, & \frac{2}{3} < x \leq 1 \end{cases}$$

Exercise 2.3. If η is constant, then

$$E(\xi|\eta) = E(\xi).$$

Proof. If $\eta \equiv C$, then the only nonempty observation event is

$$\{\eta = C\} = \Omega \implies E(\xi|\eta) = E(\xi|\Omega)\mathbb{1}_\Omega = E(\xi).$$

□

Exercise 2.4. If $0 < P(B) < 1$, then

$$E(\mathbb{1}_A|\mathbb{1}_B) = P(A|B)\mathbb{1}_B + P(A|\Omega \setminus B)\mathbb{1}_{\Omega \setminus B}.$$

Proposition 2.1. Let $\xi \in L^1$ and let η be discrete. Then $E(\xi|\eta)$ is integrable and has the following properties:

1. $E(\xi|\eta)$ is $\sigma(\eta)$ -measurable.
2. For every $A \in \sigma(\eta)$, it satisfies

$$\int_A E(\xi|\eta) dP = \int_A \xi dP.$$

Proof.

1. We write $A_n = \{\eta = y_n\}$ and set

$$c_n = \frac{\int_{A_n} \xi dP}{P(A_n)} \quad \text{and} \quad Z = \sum_{n \geq 1} c_n \mathbb{1}_{A_n} = E(\xi|\eta).$$

Each A_n belongs to $\sigma(\eta)$, so Z is $\sigma(\eta)$ -measurable.

2. It is integrable, since

$$\begin{aligned} E(|Z|) &= \sum |c_n| P(A_n) \leq \sum \int_{A_n} |\xi| dP \\ &= E(|\xi|) < \infty. \end{aligned}$$

3. Every $A \in \sigma(\eta)$ is a union $A = \bigcup_{n \in \mathcal{J}} A_n$, which implies

$$\begin{aligned} \int_A Z dP &= \sum_{n \in \mathcal{J}} c_n P(A_n) \\ &= \sum_{n \in \mathcal{J}} \int_{A_n} \xi dP \\ &= \int_A \xi dP. \end{aligned}$$

□

2.3 Conditioning on an arbitrary random variable

Definition. Let $\xi \in L^1$, and let η be any random variable. A conditional expectation of ξ given η is an integrable random variable, denoted by $E(\xi|\eta)$, with the following properties:

1. $E(\xi|\eta)$ is $\sigma(\eta)$ -measurable.
2. For every $A \in \sigma(\eta)$, it satisfies

$$\int_A E(\xi|\eta) dP = \int_A \xi dP.$$

Conditional Probability. For $A \in \mathcal{F}$, the conditional probability of A given η is defined by

$$P(A|\eta) = E(\mathbb{1}_A|\eta).$$

Lemma 2.1. Let $\mathcal{G} \subset \mathcal{F}$ be a σ -field, and let ξ be an integrable, and $\mathcal{G}$ -measurable random variable. If

$$\int_B \xi dP = 0 \quad \text{for all } B \in \mathcal{G},$$

then

$$\xi = 0 \quad \text{a.s.} \iff P(\xi \neq 0) = 0.$$

Proof. Note that

$$\{\xi \neq 0\} = \bigcup_{n \geq 1} \{|\xi| > \frac{1}{n}\} \cup \bigcup_{n \geq 1} \{|\xi| < -\frac{1}{n}\}.$$

For $\epsilon > 0$, the set $B_+ = \{\xi \geq \epsilon\}$ belongs to $\mathcal{G}$. By assumption,

$$0 = \int_{B_+} \xi dP \geq \int_{B_+} \epsilon dP = \epsilon P(B_+).$$

So, we have $P\{\xi \geq \epsilon\} = 0$. Taking $\epsilon = \frac{1}{n}$, we obtain

$$P\{\xi \neq 0\} \leq \sum_{n \geq 1} (P\{|\xi| > \frac{1}{n}\} + P\{\xi < -\frac{1}{n}\}) = 0.$$

□

Remark. Any two random variable satisfying the definition of conditional expectation are equal a.s. In particular,

$$\xi = \xi' \quad \text{a.s.} \implies E(\xi|\eta) = E(\xi'|\eta) \quad \text{a.s.}$$

Proof. Let Z, Z' satisfy the definition of conditional expectation.⁸ Then, Lemma 2.1 gives $Z = Z'$ a.s. If $\xi = \xi'$ a.s. their integral over A agree, so the same argument gives

$$E(\xi|\eta) = E(\xi'|\eta) \quad \text{a.s.}$$

□

Exercise 2.4. Take $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$, and P be the Lebesgue measure $m(\cdot)$. Define $\xi(x) = 2x^2$ and

$$\eta(x) = \begin{cases} 2, & 0 \leq x < \frac{1}{2} \\ x, & \frac{1}{2} \leq x \leq 1 \end{cases}$$

Find $E(\xi|\eta)$.

Proof. Since it is $\sigma(\eta)$ -measurable, Doob-Dynkin lemma shows that

$$E(\xi|\eta) = h(\eta) \quad \text{for some Borel function } h.$$

Since $\{\eta = 2\} = [0, \frac{1}{2})$, condition 2 gives

$$\int_{\{\eta=2\}} h(\eta) dP = \int_{\{\eta=2\}} \xi dP$$

where

$$\int_{\{\eta=2\}} h(\eta) dP = \int_0^{\frac{1}{2}} h(2) dP = \frac{1}{2} h(2) \quad \text{and} \quad \int_{\{\eta=2\}} \xi dP = \int_0^{\frac{1}{2}} 2x^2 dP = \frac{1}{12}.$$

Hence, $h(2) = \frac{1}{6}$. Next, let $B \in \mathcal{B}[\frac{1}{2}, 1]$, then $\{\eta \in B\} = B$ and condition 2 gives

$$\left(\int_B h(\eta) dP = \right) \int_B h(x) dP = \int_B 2x^2 dP (= \int_B \xi dP).$$

Since B is arbitrary, we have $h(x) = 2x^2$ a.e. in $[\frac{1}{2}, 1]$. Define

$$h(x) = \begin{cases} \frac{1}{6}, & x = 2 \\ 2x^2, & \frac{1}{2} \leq x \leq 1 \\ 0, & \text{otherwise} \end{cases} \quad \text{and} \quad Z = \begin{cases} h(\eta) = \frac{1}{6}, & 0 \leq x < \frac{1}{2} \\ h(\eta) = 2x^2, & \frac{1}{2} \leq x \leq 1 \end{cases}$$

We now verify that $Z = E(\xi|\eta)$ a.s. :

1. Since h is Borel, $Z = h(\eta)$ is $\sigma(\eta)$ -measurable.
2. Moreover, Z is bounded, so it is integrable.
3. We have already seen that

$$\int_{[0, \frac{1}{2})} Z dP = \int_{[0, \frac{1}{2})} \xi dP \quad \text{and} \quad \int_B Z dP = \int_B \xi dP \quad \text{for all } B \in \mathcal{B}[\frac{1}{2}, 1].$$

4. Every event in $\sigma(\eta)$ is either B or $[0, \frac{1}{2}) \cup B$.

⁸Note that their difference is integrable and $\sigma(\eta)$ -measurable by Proposition 1. Moreover, for all $A \in \sigma(\eta)$, we have

$$\int_A (Z - Z') dP = \int_A Z dP - \int_A Z' dP = \int_A \xi dP - \int_A \xi dP = 0.$$

Hence, by additivity of the integral, we have

$$\int_A Z dP = \int_A \xi dP \quad \text{for all } A \in \sigma(\eta).$$

□

Exercise 2.6. On $[0, 1]$ with the Lebesgue measure, let $\xi(x) = 2x^2$ and $\eta(x) = 1 - |2x - 1|$. Find $E(\xi|\eta)$.

Exercise 2.7. Let $f_{\xi, \eta}(x, y) = x + y$ on $[0, 1]^2$. Find $E(\xi|\eta)$.

Proof. We write $Z = h(\eta)$ for some Borel function h . For any $B \in \mathbb{B}$,

$$\int_{\{\eta \in B\}} Z dP = \int_{\{\eta \in B\}} \xi dP.$$

Indeed,

$$\begin{aligned} \int_{\{\eta \in B\}} Z dP &= \int_B \int_{\mathbb{R}} h(y) f_{\xi, \eta}(x, y) dx dy \\ &= \int_B h(y) \int_0^1 (x + y) dx dy \\ &= \int_B h(y) \left(\frac{1}{2} + y \right) dy \end{aligned}$$

and

$$\begin{aligned} \int_{\{\eta \in B\}} \xi dP &= \int_B \int_{\mathbb{R}} x f_{\xi, \eta}(x, y) dx dy \\ &= \int_B \int_0^1 x(x + y) dx dy \\ &= \int_B \left(\frac{1}{3} + \frac{1}{2}y \right) dy. \end{aligned}$$

Hence, we have

$$h(y) = \frac{\frac{1}{3} + \frac{1}{2}y}{\frac{1}{2} + y} = \frac{2 + 3y}{3(1 + 2y)} \quad \text{a.e. on } [0, 1].$$

We redefine

$$h(y) = \begin{cases} \frac{2+3y}{3(1+2y)}, & 0 \leq y \leq 1 \\ 0, & \text{otherwise} \end{cases}$$

□

2.4 Conditioning on a σ -field

Proposition 2.2. If $\sigma(\eta) = \sigma(\eta')$, then for all $\xi \in L^1$,

$$E(\xi|\eta) = E(\xi|\eta') \quad \text{a.s.}$$

Proof. Use Lemma 2.1

□

Definition. Let $\xi \in L^1$ and let $\mathcal{G}, \mathcal{F}$ be a σ -field. A conditional expectation of ξ given $\mathcal{G}$ is a random variable, denoted by $E(\xi|\mathcal{G})$, with the following properties:

1. $E(\xi|\mathcal{G})$ is $\mathcal{G}$ -measurable.

2. For every $A \in \mathcal{G}$, it satisfies

$$\int_A E(\xi|\mathcal{G})dP = \int_A \xi dP.$$

Remark. For $A \in \mathcal{F}$, define the conditional probability

$$P(A|\mathcal{G}) = E(\mathbb{1}_A|\mathcal{G}).$$

Concitioning on a random variable agrees with conditioning on its generated σ -field:

$$E(\xi|\eta) = E(\xi|\sigma(\eta)) \quad \text{a.s.}$$

Theorem 2.1 (Radon-Nikodym). *Let $\mathcal{G} \subset \mathcal{F}$ be a σ -field. Then for every $\xi \in L^1$, there exists an integrable, $\mathcal{G}$ -measurable random variable ζ such that*

$$\int_A \zeta dP = \int_A \xi dP \quad \text{for all } A \in \mathcal{G}.$$

Proposition 2.3. *For every $\xi \in L^1$ and $\mathcal{G} \subset \mathcal{F}$,*

$$E(\xi|\mathcal{G}) \quad \text{exists}$$

and unique up to a.s. equality.

Exercise 2.10. Show that

$$E(\xi|\{\emptyset, \Omega\}) = E(\xi) \quad \text{a.s.}$$

Proof. Let $c = E(\xi)$ and $\mathcal{G} = \{\emptyset, \Omega\}$. Then $c \in L^1$ and it is $\mathcal{G}$ -measurable. Moreover,

$$\int_{\emptyset} c dP = 0 = \int_{\emptyset} \xi dP, \quad \int_{\Omega} c dP = c = E(\xi) = \int_{\Omega} \xi dP.$$

which implies that $E(\xi|\mathcal{G}) = c$ a.s. □

Exercise 2.11. If ξ is $\mathcal{G}$ -measurable, then $E(\xi|\mathcal{G}) = \xi$ a.s.

Proof. By assumption, ξ itself is integrable and $\mathcal{G}$ -measurable. It also satisfies

$$\int_A \xi dP = \int_A \xi dP \quad \text{for all } A \in \mathcal{G}.$$

which implies that $E(\xi|\mathcal{G}) = \xi$ a.s. □

Exercise 2.12. If $B \in \mathcal{G}$, $P(B) > 0$, then

$$E(E(\xi|\mathcal{G})|B) = E(\xi|B).$$

2.5 General Properties

Proposition 2.4. *Let $\xi, \zeta \in L^1$, let $a, b \in \mathbb{R}$ and $\mathcal{H} \subset \mathcal{G} \subset \mathcal{F}$.*

1. (Linearity) $E(a\xi + b\zeta|\mathcal{G}) = aE(\xi|\mathcal{G}) + bE(\zeta|\mathcal{G})$ a.s.
2. $E(E(\xi|\mathcal{G})) = E(\xi)$ a.s.
3. If ξ is $\mathcal{G}$ -measurable, and $\xi\zeta \in L^1$, then

$$E(\xi\zeta|\mathcal{G}) = \xi E(\zeta|\mathcal{G}) \quad \text{a.s.}$$

4. If ξ is independent of $\mathcal{G}$, then

$$E(\xi|\mathcal{G}) = E(\xi) \quad a.s.$$

5. (Tower Property) $E(E(\xi|\mathcal{G})|\mathcal{H}) = E(\xi|\mathcal{H})$ a.s.

6. If $\xi \geq 0$ a.s., then $E(\xi|\mathcal{G}) \geq 0$ a.s.

Proof. Wirte

$$X = E(\xi|\mathcal{G}), \quad Z = E(\zeta|\mathcal{G}).$$

1. $aX + bZ$ is integrable and $\mathcal{G}$ -measurable. For all $A \in \mathcal{G}$, we have

$$\begin{aligned} \int_A (aX + bZ) dP &= a \int_A X dP + b \int_A Z dP \\ &= a \int_A \xi dP + b \int_A \zeta dP \\ &= \int_A (a\xi + b\zeta) dP. \end{aligned}$$

2. Taking $A = \Omega$ in the defining identity gives

$$E(X) = \int_{\Omega} X dP = \int_{\Omega} \xi dP = E(\xi).$$

6. Suppose $\xi \geq 0$ a.s. For

$$A_n = \{X \leq -\frac{1}{n}\} \in \mathcal{G},$$

the defining identity yields

$$0 \leq \int_{A_n} \xi dP = \int_{A_n} X dP \leq -\frac{1}{n} P(A_n).$$

Hence $P(A_n) = 0$ for all $n \geq 1$. Since

$$\{X < 0\} = \bigcup_{n \geq 1} A_n,$$

we have $X \geq 0$ a.e.

3. First assume

$$\xi \in \mathbb{1}_A \quad \text{where } A \in \mathcal{G}.$$

For every $B \in \mathcal{G}$,

$$\begin{aligned} \int_B \mathbb{1}_A Z dP &= \int_{B \cap A} Z dP \\ &= \int_{B \cap A} \zeta dP \\ &= \int_B \mathbb{1}_A \zeta dP. \end{aligned}$$

Now using linearity implies the result for simple $\mathcal{G}$ -measurable ξ . Then, use Dominated Convergence Theorem to extend the result to bounded $\mathcal{G}$ -measurable ξ . To consider the general case, that is, to show $\xi Z \in L^1$, let us define

$$H_n = (|\xi| \wedge n) \operatorname{sgn}(Z)$$

The bounded case and result in 2 give

$$\begin{aligned}
E((|\xi| \wedge n)|Z|) &= E(H_n Z) \\
&= E(E(H_n \zeta | \mathcal{G})) \\
&= E(H_n \zeta) \\
&\leq E(|\xi \zeta|) < \infty.
\end{aligned}$$

By Monotone Convergence Theorem, we have

$$E(|\xi Z|) \leq E(|\xi \zeta|) < \infty.$$

Now put

$$\xi_n = (-n) \vee (\xi \wedge n).$$

For every $B \in \mathcal{G}$, the bounded case gives

$$\int_B \xi_n Z dP = \int_B \xi_n \zeta dP.$$

Since $\xi_n \rightarrow \xi$ and

$$|\xi_n Z| \leq |\xi Z| \in L^1 \quad \text{and} \quad |\xi_n \zeta| \leq |\xi \zeta| \in L^1,$$

Dominated Convergence Theorem gives

$$\int_B \xi Z dP = \int_B \xi \zeta dP.$$

4. For $A \in \mathcal{G}$, independence gives

$$\begin{aligned}
\int_A E(\xi) dP &= E(\xi) P(A) \\
&= E(\xi \mathbb{1}_A) \\
&= \int_A \xi dP.
\end{aligned}$$

5. $E(E(\xi | \mathcal{G}) | \mathcal{H}) = E(\xi | \mathcal{H})$ is integrable and $\mathcal{H}$ -measurable. For every $A \in \mathcal{H} \subset \mathcal{G}$, we have

$$\int_A E(E(\xi | \mathcal{G}) | \mathcal{H}) dP = \int_A E(\xi | \mathcal{G}) dP = \int_A \xi dP$$

□

So far, we have seen that

$$\left| \int \xi dP \right| \leq \int |\xi| dP$$

our next question is whether following holds:

$$|E(\xi | \mathcal{G})| \leq E(|\xi| | \mathcal{G}) \quad \text{a.s.}$$

The notion of convexity will play a key role to prove above inequality.

Definition (Convexity). A function $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ is said to be convex if for all $x, y \in \mathbb{R}$ and $\lambda \in [0, 1]$,

$$\varphi(\lambda x + (1 - \lambda)y) \leq \lambda \varphi(x) + (1 - \lambda) \varphi(y).$$

Theorem 2.2 (Jensen's inequality). Let $\varphi : \mathbb{R} \rightarrow \mathbb{R}$ be convex. If ξ and $\varphi(\xi)$ are integrable, then

$$\varphi(E(\xi|\mathcal{G})) \leq E(\varphi(\xi)|\mathcal{G}) \quad \text{a.s.}$$

Proof. Choose a supporting affine function such that

$$a_k x + b_k \leq \varphi(x)$$

at each rational point so that

$$\varphi(x) = \sup_{k \geq 1} (a_k x + b_k).$$

For each fixed $k \geq 1$, the inequality

$$a_k \xi + b_k \leq \varphi(\xi)$$

shows that

$$a_k E(\xi|\mathcal{G}) + b_k = E(a_k \xi + b_k|\mathcal{G}) \leq E(\varphi(\xi)|\mathcal{G}) \quad \text{a.s.}$$

There are countably many rational k , so these inequalities hold simultaneously on a set of probability one, which is equivalent to holding simultaneously outside one null set. Taking their supremum over k gives

$$\varphi(E(\xi|\mathcal{G})) = \sup_{k \geq 1} (a_k E(\xi|\mathcal{G}) + b_k) \leq E(\varphi(\xi)|\mathcal{G}) \quad \text{a.s.}$$

□

Remark.

1. Since $|\cdot|$ is convex, we have

$$|E(\xi|\mathcal{G})| \leq E(|\xi||\mathcal{G}) \quad \text{a.s.}$$

which we wanted to prove.

2. If $p \geq 1$ and $\xi \in L^p$, then

$$|E(\xi|\mathcal{G})|^p \leq E(|\xi|^p|\mathcal{G}) \quad \text{a.s.}$$

which implies that

$$E(|E(\xi|\mathcal{G})|^p) \leq E(|\xi|^p).$$

3. Martingales in Discrete Time

3.1 Sequences of Random Variables

From now on, we consider ξ_n as

$$\xi_n : \Omega \rightarrow \mathbb{R}, \quad n \in \mathbb{N}.$$

Definition. Let (ξ_n) be a sequence of random variables. For a fixed outcome $\omega \in \Omega$, the sequence

$$\xi_1(\omega), \xi_2(\omega), \dots$$

is called a **sample path** of (ξ_n) .

3.2 Filtrations

Definition. A **filtration** is a sequence of σ -fields $(\mathcal{F}_n)$ such that

$$\mathcal{F}_1 \subset \mathcal{F}_2 \subset \cdots \subset \mathcal{F}.$$

Example 3.1. Let $\xi_n = 1$ for heads and $\xi_n = 0$ for tails on toss n , and put

$$\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n).$$

Let

$$A = \{\text{the first 5 tosses produce at least 2 heads}\}$$

At $n = 5$, we can decide whether A has occurred or not. But at $n = 4$, if the outcomes of the first four tosses are $TTHT$, then A remains undecided. Therefore,

$$A \in \mathcal{F}_5, \quad A \notin \mathcal{F}_4.$$

If the first four tosses are $THTH$, then it is possible to tell that A has occurred ahead at $n = 4$. However, it does not mean $A \in \mathcal{F}_4$.

Exercise 3.1. Find the smallest $n \geq 1$, if one exists, for which each of the following events belongs to $\mathcal{F}_n$.

1. $A = \{\text{the first occurrence of heads is preceded by no more than 10 tails}\}$
2. $B = \{\text{there is at least 1 head in the sequence } \xi_1, \xi_2, \dots\}$
3. $C = \{\text{the first 100 tosses produce the same outcome}\}$
4. $D = \{\text{there are no more than 2 heads and 2 tails among the first 5 tosses}\}$

Proof.

1. The event A occurs when at least one head appears among the first 11 tosses. Hence,

$$A = \bigcup_{k=1}^{11} \{\xi_k = 1\} \in \mathcal{F}_{11}.$$

After ten tails, a head on toss 11 makes A occur, whereas a tail makes it fail. Hence, $A \notin \mathcal{F}_{10}$ and the smallest n is 11.

2. After any finite string of n tails, a head may occur later or all subsequent tosses may be tails. Therefore, no finite n satisfies

$$B \in \mathcal{F}_n.$$

3. Note that

$$C = \{\xi_1 = \xi_2 = \cdots = \xi_{100} = 1\} \cup \{\xi_1 = \xi_2 = \cdots = \xi_{100} = -1\} \in \mathcal{F}_{100}.$$

After 99 heads, its occurrence still depends on toss 100. So $C \notin \mathcal{F}_{99}$.

4. $D = \emptyset \in \mathcal{F}_1$

□

Definition. A sequence (ξ_n) is **adapted to a filtration** $(\mathcal{F}_n)$ if

$$\xi_n \text{ is } \mathcal{F}_n\text{-measurable for every } n \geq 1.$$

Example 3.2. Every sequence (ξ_n) is adapted to its natural filtration $(\mathcal{F}_n)$ defined by

$$\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n).$$

Exercise 3.2. Show that

$$\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n)$$

is the smallest filtration to which (ξ_n) is adapted.

More precisely, if (ξ_n) is adapted to another filtration $(\mathcal{G}_n)$, then

$$\mathcal{F}_n \subset \mathcal{G}_n \quad \text{for all } n \in \mathbb{N}.$$

Proof. Since (ξ_n) is $\mathcal{G}_n$ -measurable, and

$$\mathcal{G}_1 \subset \mathcal{G}_2 \subset \dots \subset \mathcal{G},$$

all of $\xi_1, \dots, \xi_n$ are $\mathcal{G}_n$ -measurable. Therefore,

$$\mathcal{F}_n = \sigma(\xi_1, \dots, \xi_n) \subset \mathcal{G}_n.$$

□

3.3 Martingales

Definition (Martingale). A sequence ξ_n is a **martingale** with respect to a filtration $(\mathcal{F}_n)$ if

1. $\xi_n \in L^1$.
2. (ξ_n) is adapted to $(\mathcal{F}_n)$.
3. $E(\xi_{n+1} | \mathcal{F}_n) = \xi_n$ a.s.

Example 3.3. Let (η_n) be independent random variables with $E(\eta_n) = 0$ for every $n \geq 1$. Define

$$\xi_n = \sum_{k=1}^n \eta_k, \quad \mathcal{F}_n = \sigma(\eta_1, \dots, \eta_n).$$

Then (ξ_n) is a martingale with respect to $(\mathcal{F}_n)$.

Proof. The sum ξ_n is $\mathcal{F}_n$ -measurable and integrable, since

$$E(|\xi_n|) \leq \sum_{k=1}^n E(|\eta_k|) < \infty.$$

Since η_{n+1} is independent of $\mathcal{F}_n$, we have

$$\begin{aligned} E(\xi_{n+1} | \mathcal{F}_n) &= E(\xi_n + \eta_{n+1} | \mathcal{F}_n) \\ &= E(\xi_n | \mathcal{F}_n) + E(\eta_{n+1} | \mathcal{F}_n) \\ &= \xi_n + E(\eta_{n+1}) = \xi_n. \end{aligned}$$

□

Example 3.4. Let $\xi \in L^1$, and let $(\mathcal{F}_n)$ be a filtration. Then the sequence defined by

$$\xi_n = E(\xi | \mathcal{F}_n)$$

is a martingale with respect to $(\mathcal{F}_n)$.

Proof. ξ_n is integrable, $\mathcal{F}_n$ measurable by definition of conditional expectation. Furthermore, since $\mathcal{F}_n \subset \mathcal{F}_{n+1}$, we have

$$E(\xi_{n+1}|\mathcal{F}_n) = E(E(\xi|\mathcal{F}_{n+1})|\mathcal{F}_n) = E(\xi|\mathcal{F}_n) = \xi_n.$$

□

Exercise 3.3. Show that a martingale (ξ_n) has constant expectation:

$$E(\xi_n) = E(\xi_1) \quad \text{for all } n \geq 1.$$

Proof. By definition of martingale and conditional expectation, we have

$$E(\xi_{n+1}) = E(E(\xi_{n+1}|\mathcal{F}_n)) = E(\xi_n).$$

By induction, we have $E(\xi_n) = E(\xi_1)$ for all $n \geq 1$.

□

Exercise 3.4. Suppose (ξ_n) is a martingale with respect to $(\mathcal{F}_n)$. Show that it is also martingale with respect to its natural filtration $(\mathcal{G}_n)$ defined by

$$\mathcal{G}_n = \sigma(\xi_1, \dots, \xi_n).$$

Proof. Since $\mathcal{G}_n \subset \mathcal{F}_n$, we have

$$\begin{aligned} E(\xi_{n+1}|\mathcal{G}_n) &= E(E(\xi_{n+1}|\mathcal{F}_n)|\mathcal{G}_n) \\ &= E(\xi_n|\mathcal{G}_n) = \xi_n. \end{aligned}$$

□

Exercise 3.5 (Symmetric Random Walk). Let (η_n) be independent random variables with

$$P\{\eta_n = 1\} = P\{\eta_n = -1\} = \frac{1}{2},$$

and define the symmetric random walk by

$$\xi_0 = 0, \quad \xi_n = \sum_{k=1}^n \eta_k, \quad n \geq 1.$$

Show that $\xi_n^2 - n$ is a martingale for the filtration

$$\mathcal{F}_n = \sigma(\eta_1, \dots, \eta_n), \quad \mathcal{F}_0 = \{\emptyset, \Omega\}.$$

Proof. Since $|\xi_n| \leq n$, the variable $\xi_n^2 - n$ is integrable, and it is $\mathcal{F}_n$ -measurable. The new step is independent of $\mathcal{F}_n$ and satisfies

$$E(\eta_{n+1}) = 0 \quad \text{and} \quad E(\eta_{n+1}^2) = 1.$$

Thus,

$$\begin{aligned} E(\xi_{n+1}^2 - (n+1)|\mathcal{F}_n) &= E((\xi_n + \eta_{n+1})^2 - (n+1)|\mathcal{F}_n) \\ &= E(\xi_n^2 + 2\xi_n\eta_{n+1} + \eta_{n+1}^2|\mathcal{F}_n) - (n+1) \\ &= \xi_n^2 + 2\xi_n E(\eta_{n+1}) + E(\eta_{n+1}^2) - (n+1) \\ &= \xi_n^2 - n. \end{aligned}$$

□

Exercise 3.6. $\zeta_n = (-1)^n \cos(\pi \xi_n)$ is a martingale.

Proof. Homework. □

Definition. A sequence (ξ_n) is a **submartingale** with respect to a filtration $(\mathcal{F}_n)$ if

1. $\xi_n \in L^1$.
2. (ξ_n) is adapted to $(\mathcal{F}_n)$.
3. $E(\xi_{n+1}|\mathcal{F}_n) \geq \xi_n$ a.s.

Similarly, (ξ_n) is a **supermartingale** with respect to $(\mathcal{F}_n)$ if

1. $\xi_n \in L^1$.
2. (ξ_n) is adapted to $(\mathcal{F}_n)$.
3. $E(\xi_{n+1}|\mathcal{F}_n) \leq \xi_n$ a.s.

Exercise 3.7. Suppose (ξ_n) is a martingale⁹ with $\xi_1 \in L^2$. Show that (ξ_n^2) is a submartingale with respect to the same filtration.

Proof. Since $\xi_n \in L^2$, its square is integrable. It is clear that

$$\xi_n^2 \text{ is } \mathcal{F}_n\text{-measurable for every } n \geq 1.$$

Conditional Jensen, applied to $x \mapsto x^2$, gives

$$E(\xi_{n+1}^2|\mathcal{F}_n) \geq (E(\xi_{n+1}|\mathcal{F}_n))^2 = \xi_n^2.$$

□

Indeed, the result of Exercise 3.7 can be generalized to any convex function φ . That, is, if $\varphi(\xi_n) \in L^1$, then $(\varphi(\xi_n))$ is a submartingale.

3.4 Games of Chance

Definition. A sequence (α_n) is **previsible** with respect to a filtration $(\mathcal{F}_n)$ if α_n is $\mathcal{F}_{n-1}$ -measurable for every $n \geq 1$.

In gambling, α_n is the stake for round n , and (α_n) is called a **gambling strategy**. Let us consider following example:

1. (η_n) : winnings per unit stake in game n .
2. $\xi_n = \eta_1 + \dots + \eta_n$: total winnings after n games with $\xi_0 = 0$.
3. $\mathcal{F}_n := \sigma(\eta_1, \dots, \eta_n)$, with $\mathcal{F}_0 = \{\emptyset, \Omega\}$.

The game is said to be ...

1. **fair** if $E(\xi_n|\mathcal{F}_{n-1}) = \xi_{n-1}$ a.s.
2. **favorable** if $E(\xi_n|\mathcal{F}_{n-1}) \geq \xi_{n-1}$ a.s.
3. **unfavorable** if $E(\xi_n|\mathcal{F}_{n-1}) \leq \xi_{n-1}$ a.s.

⁹From now on, we consider that martingale (ξ_n) is adapted to its natural filtration $(\mathcal{F}_n)$.

Suppose you may vary your stake from game to game. Under a strategy (α_n) , your total winnings are

$$\zeta_0 = 0, \quad \zeta_n = \sum_{k=1}^n \alpha_k (\xi_k - \xi_{k-1}).$$

Proposition 3.1. *Let (α_n) be previsible strategy such that each α_n is bounded and let (ζ_n) be the winnings defined above. If (ξ_n) is a martingale, then (ζ_n) is a martingale for the same filtration. If also $\alpha \geq 0$ for every n , then a supermartingale (ξ_n) yields supermartingale (ζ_n) , and a submartingale (ξ_n) yields a submartingale (ζ_n) .*

Proof. Each term in ζ_n is $\mathcal{F}_n$ -measurable. Boundedness of the stakes also gives

$$E(|\alpha_k(\xi_k - \xi_{k-1})|) \leq \|\alpha_k\|_\infty E(|\xi_k - \xi_{k-1}|) < \infty.$$

So every ζ_n is integrable.

Since α_n and ζ_{n-1} are $\mathcal{F}_{n-1}$ -measurable, the identity

$$\zeta_n = \zeta_{n-1} + \alpha_n(\xi_n - \xi_{n-1})$$

implies

$$\begin{aligned} E(\zeta_n | \mathcal{F}_{n-1}) &= \zeta_{n-1} + \alpha_n(E(\xi_n | \mathcal{F}_{n-1}) - \xi_{n-1}) \\ &= \zeta_{n-1}. \end{aligned}$$

□

Appendix

A. Measure Theory

A.1 Measures and Measurable Functions